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ROLL NO : 22CSB07

DEPT : CSE

SUBJECT : Engineering Mathematics - II (M2)

PART A

1. A first order differential equation is an equation that contains the first derivative of a function but doesn't contain higher derivatives. Example : $\frac{dy}{dx} + y = x$

2. Differential Equation is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

3. Linear differential equation of first order is written in the form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where $P(x)$ and $Q(x)$ are function of x

4. The Laplace transform of a function $f(t)$ is defined as

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$$

It converts a function from the time domain into s-domain.

5. The rank of a matrix is the maximum number of linearly independent rows or columns in the matrix

PART B

6. $\frac{dy}{dx} + y = e^x \longrightarrow (1)$

Standard form: $\frac{dy}{dx} + P(x)y = Q(x)$

here, $P(x) = 1$

$Q(x) = e^x$

Integrating Factor (IF) $= e^{\int P(x)dx}$

$= e^{\int 1 dx}$

$IF = e^x \longrightarrow (2)$

multiply (1) by (2)

$$e^x \frac{dy}{dx} + e^x y = e^{2x}$$

differentiate the equation

$$\frac{dye^x}{dx} = e^x \frac{dy}{dx} + e^x y$$

$$\frac{d(ye^x)}{dx} = e^{2x}$$

Integrating both sides

$$\int \frac{d}{dx}(ye^x)\,dx = \int e^{2x}dx$$

$$ye^x = \frac{e^{2x}}{2} + c \quad \text{[divide by } e^x\text{]}$$

$$\boxed{y = \frac{e^x}{2} + c}$$

7 $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$

$M = 2xy + y^2$

$N = x^2 + 2xy$

Condition for exactness $\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$$\frac{\partial M}{\partial y} = 2x + 2y \qquad \frac{\partial N}{\partial x} = 2x + 2y$$

As both are equal, the equation is exact

Integrating M with respect to $x \Rightarrow \int (2xy + y^2)\,dx$

$\Rightarrow x^2y + xy^2 + h(y)$ assume y as constant

Differentiate w.r.t $y \Rightarrow \frac{d}{dy}(x^2y + xy^2 + h(y))$

$= x^2 + 2xy + h'(y)$ [compare with N]

$h'(y) = 0$

$h(y) = C$

$= x^2y + xy^2 = C$

8. $f(t) = te^{at}$

Formula used $\Rightarrow \frac{1}{(s-a)^2}$

$$L\{te^{at}\} = \frac{1}{(s-a)^2}$$

Laplace transform of function $f(t)$ is defined as

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$$

which helps converts a function from the time domain into the s-domain

9. $F(s) = \frac{s+1}{(s+1)^2 + 4}$

standard formula:

$$L\{e^{-at}\cos bt\} = \frac{s+a}{(s+a)^2 + b^2}$$

So according to the formula, in the given equation

$a = 1$
$b = 2$

$$L^{-1}\left\{\frac{s+1}{(s+1)^2+4}\right\} = e^{-1st}\cos(2)t$$

$$= e^{-t}\cos 2t$$

10. $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

Formula : $|A - \lambda I| = 0$

$$\left|\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right| = 0$$

$$\left|\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right| = 0$$

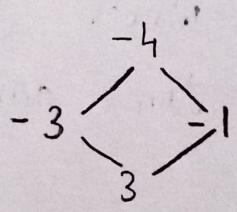
$$\begin{vmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda)^2 - 1 = 0$$

$$4 - 4\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$\lambda(\lambda-3) - 1(\lambda-3) = 0$$

$$(\lambda-3)(\lambda-1) = 0$$

$$\boxed{\lambda = 3 \quad , \lambda = 1}$$